

Bandung Institute of Technology

Informatics / School of Electrical Engineering and Informatics

IF5251 - AI in Production

Electricity Demand Forecasting



Group 04

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Academic Year 2025/2026 - Semester 2

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Chapter 1

Requirements

Electricity demand forecasting is a critical component of modern energy grid management. Accurate demand forecasts allow grid operators to balance supply and load proactively, reduce reliance on expensive peaking generation capacity, and facilitate the integration of variable renewable energy sources. This chapter defines the requirements for an AI-based electricity demand forecasting system that operates across multiple geographic locations, covering its goals, the needs of its intended users, the assumptions about the operating environment, and the metrics used to assess both model and system performance.

1.1 System and Model Goals

The *system goal* is to provide **accurate, automated, and near-real-time electricity demand forecasts** for multiple geographic locations, enabling energy operators and planners to make informed grid management decisions. The system ingests historical hourly consumption data alongside meteorological and temporal inputs spanning from 2016 onwards, and exposes forecast results through a web-based interface.

The *model goal* is to learn a **regression function** that maps a rich feature set which consists of over 30 meteorological variables. These features span from weather features such as temperature, humidity, solar radiation, and wind speed, to location identifiers and temporal features. The target variable, y , represents hourly electricity consumption per location cluster. Concretely, the model is expected to perform the following tasks.

- **Forecast** hourly electricity consumption for up to 24 hours ahead for each monitored location.
- **Generalise** across climatically and geographically diverse locations (e.g. London UK, Oakland CA, Groningen NL, Ottawa CA, Washington DC as the locations currently available).
- **Capture** non-linear relationships between weather drivers and demand, including seasonal patterns, diurnal cycles, and weather extremes.
- **Remain performant** on unseen future data through periodic retraining as new consumption records accumulate.

1.2 User Requirements

Based on domain analysis, the following user requirements have been identified in Table 1.1. These requirements are made on the basis of the intended use cases for the forecasting system, which include operational grid management, short-term energy procurement, and demand response planning.

In addition to that, these requirements assume that the users have a certain level of domain expertise in energy management and are familiar with interpreting time-series forecasts in the context of grid operations.

1.2.1 User Types

The system targets two primary user types: **energy grid operators**, who rely on short-term forecasts for real-time grid balancing decisions, and **energy planning analysts**, who use forecasts to support procurement and demand response planning. Both groups require domain expertise in energy management and are assumed to be comfortable interpreting time-series forecasts in an operational context.

The system is **not intended** for end consumers or non-technical stakeholders. Table 1.2 maps each user requirement to the applicable user type, along with representative usage scenarios and implementation priority.

1.2.2 Use Cases

Based on the user requirements defined earlier, the following use cases have been identified for the electricity demand forecasting system, as illustrated in Figure 1.1.

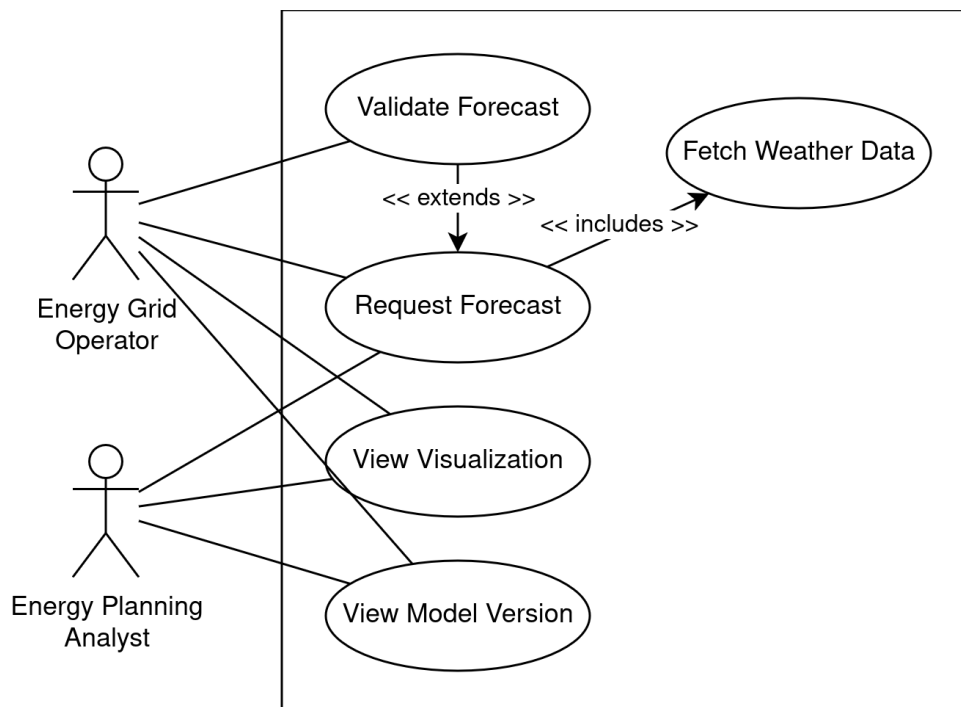


Figure 1.1: System Use Case Diagram

Each use case corresponds to a specific interaction between the user and the system, aligned with the requirements outlined in Table 1.1, Table 1.3 explains this further.

Table 1.3: Use Cases Overview

ID	Name	Description	Input	Output	Related Req.
UC-01	Request Forecast	User selects a location and time horizon to request a demand forecast via the web interface.	Location identifier, time horizon (up to 48 h)	Inference forecast result	UR-01, UR-02, UR-03, UR-07, UR-09
UC-02	View Visualisation	User views forecast results as a labelled hourly time-series plot.	Inference forecast result	Time-series plot rendered in the interface	UR-04
UC-03	Validate Forecast	User flags a low-confidence forecast for manual review. Extends UC-01.	Forecast result	Correction queued for retraining	UR-05, UR-08
UC-04	View Model Version	User checks which model version produced a given forecast.	Forecast identifier	Model version and training date	UR-06
UC-05	Fetch Weather Data	System automatically retrieves meteorological data for the requested location. Included in UC-01.	Location identifier, request timestamp	Variable meteorological feature set	UR-03

1.3 Environment Assumptions

Environment assumptions explicitly define the conditions under which the system is expected to operate correctly. They bound the system's scope and serve as the basis for design decisions made in subsequent chapters. The assumptions are listed in Table 1.4.

Deviations from these assumptions may require system or model updates, and any new assumptions that arise during development should trigger a reassessment of the relevant requirements.

1.4 System Components and Interactions

This section identifies the AI and non-AI components that constitute the forecasting system, maps their interactions, and describes the feedback loop through which the system improves over time.

1.4.1 AI and Non-AI Components

Table 1.5 categorises the system components into AI and non-AI elements along with their roles.

Figure 1.2 illustrates the interactions between system components. In the diagram, arrows indicate the direction of data or request flow between components. A dashed arrow represents logging, where data is written passively as a side effect of an operation rather than as a primary request. External components are shown with a dashed border.

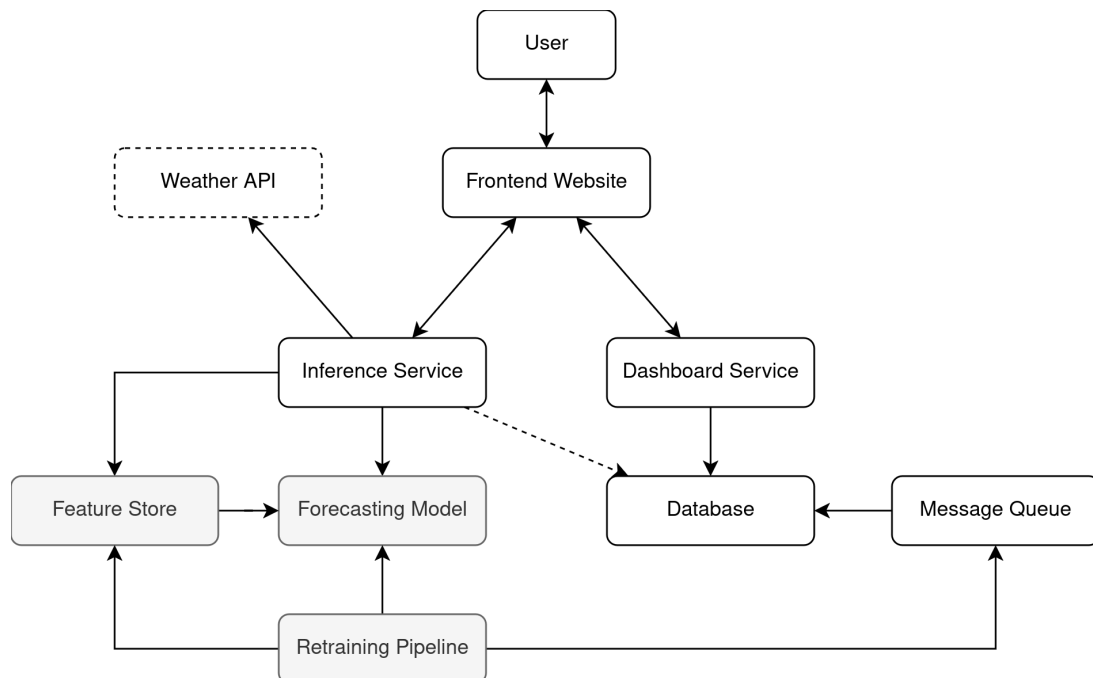


Figure 1.2: System Components and Interactions Diagram

The user interacts exclusively with the *Frontend Website*, which routes forecast requests to the *Inference Service* and historical data queries to the *Dashboard Service*. The *Inference Service* fetches preprocessed feature vectors from the *Feature Store* and passes them to the *Forecasting Model* to produce a forecast result. Weather data is retrieved from the external *Weather API* and routed through the *Inference Service* before being stored in the *Feature Store*. Forecast results and operator validations are logged to the *Database* via the *Inference Service*. The *Dashboard Service* exposes historical consumption and weather records from the *Database* to the *Frontend Website*. Validated corrections flow from the *Database* to the *Message Queue*, which triggers the *Retraining Pipeline* once the threshold is reached. The *Retraining Pipeline* pulls features from the *Feature Store* and updates the *Forecasting Model* with the newly trained weights.

1.4.2 Feedback Loop

The system incorporates an **operator correction loop** that enables continuous model improvement. Figure 1.3 illustrates this feedback mechanism.

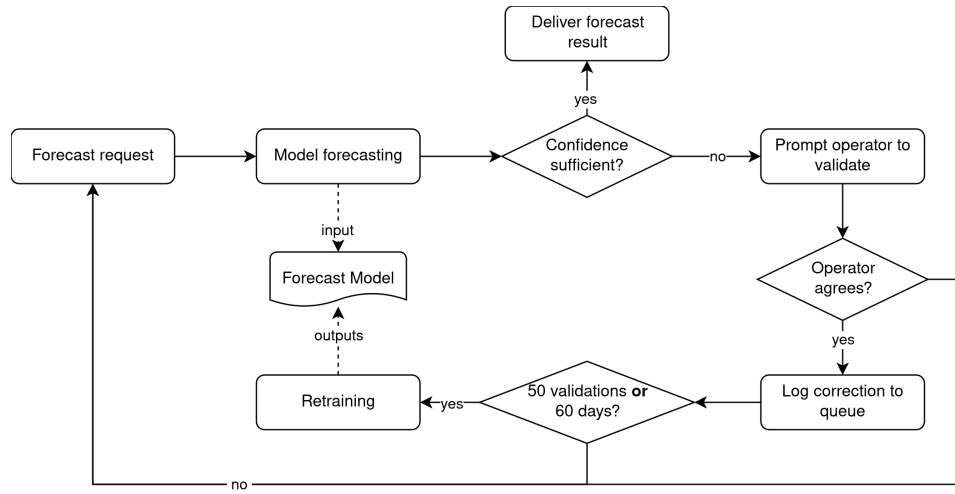


Figure 1.3: System Feedback Loop

When a forecast request is received, the model produces a result which is then checked against a confidence threshold. If confidence is sufficient, the forecast is delivered directly to the operator. Otherwise, the operator is prompted to validate the forecast.

If the operator agrees, the correction is logged to the validation queue; if not, the forecast is dismissed without further action. Once the queue reaches 50 validated corrections, or 60 days have elapsed since the last retraining, whichever comes first, a retraining job is triggered. The retrained model then replaces the previous one and resumes serving forecast requests.

1.5 Metrics for Measurement

Measuring system performance requires a combination of model-level (functional) and system-level (non-functional) metrics. Functional metrics assess the accuracy and reliability of the demand forecasts produced by the regression model. Non-functional metrics assess the operational characteristics of the deployed service, including responsiveness, scalability, and availability.

1.5.1 Functional Metrics

The following functional metrics are used to evaluate the regression model's forecasting accuracy. All metrics are computed on a held-out test set comprising data from a time period not seen during training, across all monitored locations. For each metric, specific target thresholds are defined based on domain standards and operational requirements for electricity demand forecasting, which are shown in Table 1.6.

- **MAE** (Mean Absolute Error): the average absolute deviation between predicted and actual consumption; interpretable in the same unit as the target variable and robust to outliers.

- **RMSE** (Root Mean Squared Error): penalises large forecast errors more heavily than MAE, making it sensitive to extreme deviations that are operationally costly.
- **MAPE** (Mean Absolute Percentage Error): expresses error as a percentage of actual consumption, enabling fair comparison across locations with different absolute demand scales.
- **R²** (Coefficient of Determination): measures the proportion of variance in the target variable explained by the model, where a value of 1.0 indicates a perfect fit.

1.5.2 Non-Functional Metrics

Non-functional metrics govern the quality of the deployed service independent of model accuracy. These metrics ensure the system is operationally reliable for production use by energy grid operators operating on a 24/7 basis. For each non-functional metric, specific target thresholds are defined based on the operational requirements of real-time forecasting services in the energy sector, which are shown in Table 1.7.

- **Latency**: the end-to-end response time from API request submission to forecast delivery. Low latency is critical in grid management contexts where decisions are time-sensitive.
- **Throughput**: the number of concurrent forecast requests the service can handle per unit time without degradation, ensuring the system scales to multiple simultaneous operator queries.
- **Availability**: the percentage of time the forecasting service is accessible. Unplanned downtime has direct operational consequences for grid operators who depend on continuous access.
- **Model Freshness**: the maximum permissible age of the deployed model before retraining is mandated, ensuring that concept drift and seasonal pattern shifts do not degrade forecast accuracy over time.

Table 1.7: Non-Functional Metrics Overview

Metric	Target
Latency	< 500 ms
Throughput	≥ 50 req/s
Availability	$\geq 99.5\%$
Model Freshness	≤ 60 days

1.6 Failure Mode and Effects Analysis

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Figure 1.4: FMEA Diagram Placeholder

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Table 1.1: User Requirements Overview

ID	Requirement Name	Description	Rationale
UR-01	Forecast Query	Users shall be able to request an electricity demand forecast for a specific monitored location and a time horizon of up to 48 hours via a web-based interface.	Enables users to obtain actionable forecasts for operational decision-making.
UR-02	Location Selection	The system shall support all locations present in the training dataset, selectable by name or geographic identifier (e.g. city name or <code>location_id</code>).	Allows users to access forecasts relevant to their specific geographic area of interest.
UR-03	Weather Data Fetch	The system shall fetch the current relevant weather data for each location automatically.	Reduces user burden by eliminating manual weather input, while ensuring forecasts are based on current meteorological data.
UR-04	Forecast Visualisation	Forecast results shall be displayed as a time-series plot of predicted hourly consumption over the selected time window, with clear axis labelling.	Facilitates intuitive understanding of forecast trends and patterns.
UR-05	Validation Interface	Users must be able to flag or manually validate forecasts that fall below a confidence threshold.	Allows users to ensure forecast quality and accuracy and supplying additional data for retraining.
UR-06	Model Version Visibility	Users must be able to see which version of the forecasting model was used to generate a particular forecast.	Enables users to understand the context and potential staleness of each forecast.
UR-07	Response Timeliness	Forecast results shall be returned within an operationally acceptable response time to support real-time grid management decisions.	Ensures forecasts are delivered in a timeframe that allows for timely action.
UR-08	Accuracy Transparency	The interface shall communicate expected model error ranges so users can appropriately calibrate their trust in the forecasts.	Builds user trust by setting realistic expectations about forecast reliability.
UR-09	Continuous Availability	The forecasting service shall be accessible at all times, reflecting the 24/7 nature of grid operations.	Ensures users can access forecasts whenever needed without interruption.

Table 1.2: User Requirements Mapping to System Users

ID	Energy Grid Operator	Energy Planning Analyst	Usage Scenario	Priority
UR-01	✓	✓	Requesting forecasts for operational decision-making.	High
UR-02	✓	✓	Selecting specific locations for forecast retrieval.	High
UR-03	✓	✓	System automatically fetches weather data upon location selection.	High
UR-04	✓	✓	Visualising forecast results for interpretation.	Medium
UR-05	✓	✗	Validating forecasts that fall below confidence thresholds.	Medium
UR-06	✓	✓	Seeing which version of the model was used.	Medium
UR-07	✓	✗	Ensuring forecasts are delivered in a timely manner for real-time actions.	High
UR-08	✓	✓	Understanding expected error ranges to calibrate trust.	Medium
UR-09	✓	✓	Accessing forecasts at any time without interruption.	High

Table 1.4: Environment Assumptions Overview

ID	Assumption Name	Description
EA-01	Weather Data Availability	Real-time and forecast meteorological data matching the 30+ feature schema used during training is continuously available from the Weather API as model input at inference time.
EA-02	Hourly Temporal Resolution	All historical and incoming data is timestamped at hourly granularity. Sub-hourly or irregular inputs are outside the current scope.
EA-03	Stationarity of Demand Patterns	The relationship between weather conditions, time features, and electricity consumption is assumed to be sufficiently stable over time for the trained model to remain valid between retraining cycles. Major structural disruptions may require earlier retraining.
EA-04	Multi-Location Heterogeneity	Each location has a distinct demand profile driven by local climate, building stock, and population density. The <code>location_id</code> and <code>cluster_size</code> fields capture this heterogeneity and must be present at inference time.
EA-05	Data Quality	Historical data is assumed to have been preprocessed (null values handled, outliers addressed) as reflected in the <code>full_data_non_null.csv</code> dataset. Incoming data at inference time is expected to pass basic schema and range validation checks before being fed to the model.
EA-06	Deployment Infrastructure	The system is deployed on a server or cloud instance capable of hosting the trained model and serving REST API requests from the Next.js frontend. Internal component communication is assumed to be stable enough to meet the 500 ms latency target.
EA-07	Weather API Reliability	The external weather data provider is assumed to maintain sufficient uptime and data consistency. Prolonged API unavailability or schema changes may require fallback handling or data pipeline updates.
EA-08	Location Scope	The system only serves locations present in the training dataset. Queries for out-of-distribution locations are outside the current scope and will not produce valid forecasts.
EA-09	Access Control	The system is assumed to be deployed in a controlled environment accessible only to authorized personnel. No public-facing authentication layer is required within the current scope.

Table 1.5: System Components Overview

Type	Component	Role
AI	Forecasting Model	Produces hourly electricity demand forecasts and confidence estimates.
Non-AI	Inference Service	Serves model inference requests, fetches preprocessed features from the Feature Store and current weather data from the Weather API, and routes forecast results to the frontend.
Non-AI	Retraining Pipeline	Consumes validated corrections from the Message Queue, retrieves features from the Feature Store, re-trains the Forecasting Model, and registers the updated model for serving.
Non-AI	Feature Store	Manages and serves consistent feature sets for both training and inference.
Non-AI	Database	Persists historical consumption data, forecast records, and validation logs.
Non-AI	Message Queue	Buffers operator validations and triggers the Retraining Pipeline when the queue threshold is reached.
Non-AI	Dashboard Service	Exposes a read API for the frontend to query historical weather and electricity consumption records from the database.
Non-AI	Frontend Website	Provides the web interface for forecast queries, visualisation, and validation prompts.
External	Weather API	Provides real-time and forecast meteorological data for the requested location at inference time.

Table 1.6: Functional Metrics Overview

Metric	Description	Target
MAE	Mean Absolute Error	< 10% of mean target per location
RMSE	Root Mean Squared Error	< 15% of mean target per location
MAPE	Mean Absolute Percentage Error	< 10%
R ²	Coefficient of Determination	> 0.85

Chapter 2

Architecture and Design

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2.1 Modelling Tradeoffs

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2.2 Deployment Architecture

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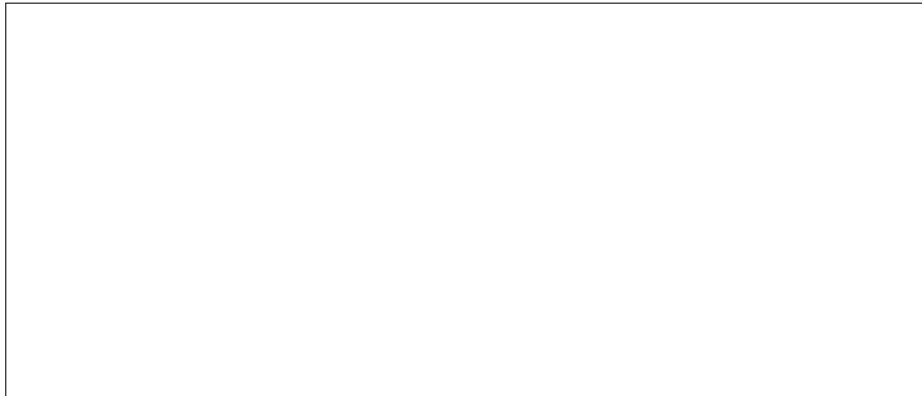


Figure 2.1: Deployment Architecture Diagram Placeholder

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2.3 Data Science Pipeline

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Figure 2.2: Data Science Pipeline Diagram Placeholder

Nulla in ipsum. Praesent eros nulla, congue vitae, euismod ut, commodo a, wisi.

2.4 Monitoring

Model monitoring ensures that the Forecasting Model continues to perform within acceptable bounds after deployment. The system uses Evidently to track model behaviour in production across two dimensions: data quality and prediction drift.

Data Quality Monitoring Evidently monitors incoming inference requests for schema violations, out-of-range feature values, and missing fields against the expected 30+ variable meteorological feature schema. This directly supports environment assumption EA-05, which requires incoming data to pass basic schema and range validation before being fed to the model. Any violations are logged to the Database and trigger an alert to the responsible personnel.

Prediction Drift Monitoring After each forecast window elapses, actual consumption records are ingested and compared against prior forecasts. Evidently computes drift metrics over a rolling window to detect whether the distribution of forecast errors has shifted significantly from the baseline established at training time. If drift is detected, an alert is raised and the event is logged. This connects directly to the hybrid retraining trigger described in Section ??, sustained drift is one of the signals that may prompt earlier retraining outside the scheduled 60-day cycle.

Monitoring Dashboard Evidently generates monitoring reports that are surfaced through the Grafana dashboard, giving operators visibility into model health over time. Key indicators displayed include feature distribution shifts, rolling MAPE, and the current validation queue size relative to the retraining threshold.

2.5 Anticipate Evolution

The system is designed with extensibility in mind, recognising that operational requirements and data availability will evolve over time. The following scenarios represent anticipated directions of change and the mechanisms through which the system accommodates them.

New Locations The system currently supports five monitored locations. Adding new locations requires sourcing historical consumption data for the target city, registering the corresponding `location_id` and `cluster_size` entries in the Feature Store, and retraining or fine-tuning the Forecasting Model to cover the new demand profile. Model performance on the new location must be validated against the functional metric targets defined in Section ?? before the location is made available to users.

New Weather Features If the Weather API introduces additional meteorological variables, or the team identifies new signals that improve forecast accuracy, the feature schema must be updated consistently across the Feature Store, the model input layer, and the data validation pipeline. Environment assumption EA-01 would require reassessment to reflect the updated schema, and the model must be retrained on the expanded feature set before deployment.

Scaling to More Users The current throughput target of 50 req/s is sufficient for the anticipated professional user base. Should usage grow beyond this threshold, Kubernetes horizontal pod autoscaling allows additional Inference Service replicas to be provisioned without architectural changes. The stateless design of the Inference Service ensures that new replicas can handle requests independently without shared in-memory state.

Retraining Threshold Tuning The hybrid retraining trigger, 50 validated corrections or 60 days, whichever comes first, is an initial estimate based on expected validation frequency during the assignment period. In production, these thresholds may require adjustment as actual validation patterns become clearer. Both parameters should be exposed as configurable values in the system configuration rather than hardcoded, allowing tuning without redeployment.

2.6 Big Data Processing

Donec et nisl at wisi luctus bibendum. Nam interdum tellus ac libero. Sed sem justo, laoreet vitae, fringilla at, adipiscing ut, nibh. Maecenas non sem quis tortor eleifend fermentum.

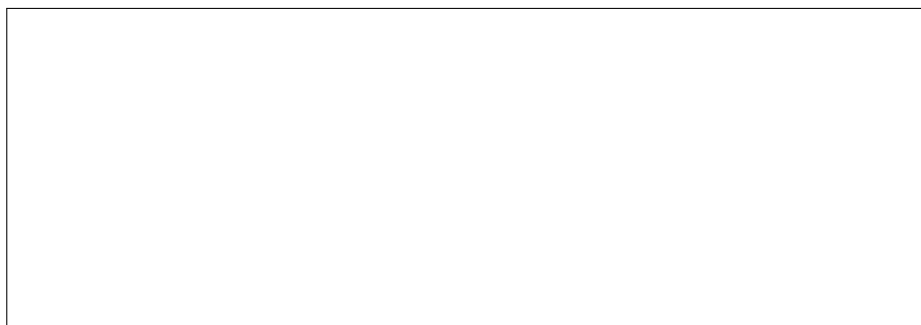


Figure 2.3: Big Data Processing Architecture Placeholder

Nulla non mauris vitae wisi posuere convallis. Sed eu nulla nec eros scelerisque pharetra.

2.7 Human-AI Design

Nulla ac nisl. Nullam urna nulla, ullamcorper in, interdum sit amet, gravida ut, risus. Aenean ac enim. In luctus.

Chapter 3

Quality Assurance

Etiam pede massa, dapibus vitae, rhoncus in, placerat posuere, odio. Vestibulum luctus commodo lacus. Morbi lacus dui, tempor sed, euismod eget, condimentum at, tortor.

3.1 Model Testing

Etiam suscipit aliquam arcu. Aliquam sit amet est ac purus bibendum congue. Sed in eros. Morbi non orci.

Table 3.1: Model Testing Results Placeholder

Test Case	Expected Output	Actual Output
Case 1	Placeholder	Placeholder
Case 2	Placeholder	Placeholder
Case 3	Placeholder	Placeholder

3.2 Data Quality

Donec et nisl id sapien blandit mattis. Aenean dictum odio sit amet risus. Morbi purus. Nulla a est sit amet purus venenatis iaculis.

3.3 QA Automation

Maecenas non massa. Vestibulum pharetra nulla at lorem. Duis quis quam id lacus dapibus interdum. Nulla lorem.

3.4 Testing in Production

Vivamus eu tellus sed tellus consequat suscipit. Nam orci orci, malesuada id, gravida nec, ultricies vitae, erat. Donec risus turpis, luctus sit amet, interdum quis, porta sed, ipsum.

Adversarial Attacks

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Load Testing

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3.5 Infrastructure Quality

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3.6 Debugging

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Chapter 4

Operations

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4.1 Unit Testing, Integration Testing and Continuous Training Pipeline

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Figure 4.1: Continuous Training Pipeline Diagram Placeholder

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4.2 Configuration Management

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4.3 Monitoring

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Figure 4.2: Monitoring Dashboard Placeholder

Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Aenean nonummy turpis id odio.

4.4 Versioning

Nulla malesuada risus ut urna. Aenean pretium velit sit amet metus. Duis iaculis. In hac habitasse platea dictumst.

4.5 MLOps

Donec tempus neque vitae est. Aenean egestas odio sed risus ullamcorper ullamcorper. Sed in nulla a tortor tincidunt egestas. Nam sapien tortor, elementum sit amet, aliquam in, porttitor faucibus, enim.

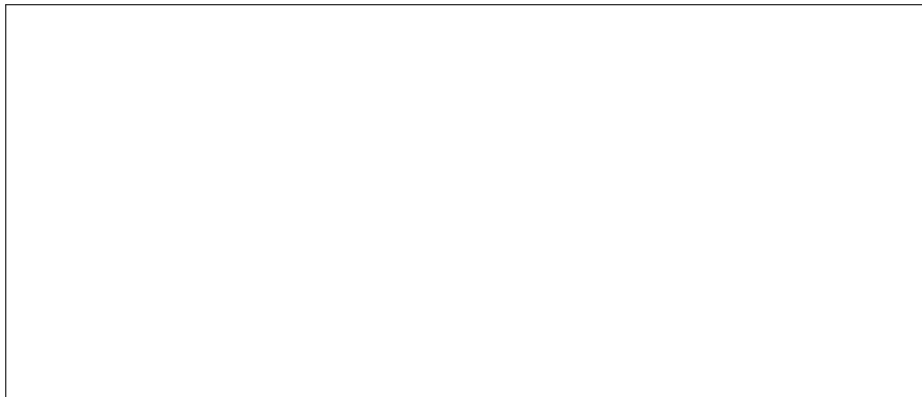


Figure 4.3: MLOps Workflow Diagram Placeholder

Fusce suscipit cursus sem. Vivamus risus mi, egestas ac, imperdiet varius, faucibus quis, leo.

Chapter 5

Responsible AI

Praesent sed neque id pede mollis rutrum. Vestibulum iaculis risus. Pellentesque lacus.

5.1 Privacy

Sed gravida lectus ut purus. Morbi laoreet magna. Pellentesque eu wisi. Proin turpis.

5.2 Sensitive Audits

Curabitur ac lorem. Vivamus non justo in dui mattis posuere. Etiam accumsan ligula id pede. Maecenas tincidunt diam nec velit.

Table 5.1: Sensitive Audit Checklist Placeholder

Audit Item	Status	Notes
Data Anonymisation	Pending	Placeholder
Bias Assessment	Pending	Placeholder
Fairness Evaluation	Pending	Placeholder

5.3 Explainability

Quisque consectetur. In suscipit mauris a dolor pellentesque consectetur. Mauris convallis neque non erat. In lacinia.



Figure 5.1: Model Explainability (SHAP / LIME) Placeholder

Maecenas accumsan dapibus sapien. Duis pretium iaculis arcu.

5.4 Transparency

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